

Solutions to Quiz 1

Q 1

Given a polynomial $f(x) = \sum_{k=0}^m c_k x^k$ with integer coefficients $c_k \in \mathbb{N}$.

Suppose $f(a)$ is divisible by N for some integers $a, N \in \mathbb{N}$.

Show that $f(a + b \cdot N)$ is divisible by N for every integer b .

Solution: We note that

$$f(a + b \cdot N) = \sum_{k=0}^m c_k (a + b \cdot N)^k = \sum_{k=0}^m c_k \sum_{r=0}^k \binom{k}{r} a^r b^{k-r} N^{k-r}$$

Note that for each k

$$\sum_{r=0}^k \binom{k}{r} a^{k-r} b^r N^r = a^k + b_k \cdot N$$

where $b_k = \sum_{r=1}^k \binom{k}{r} a^{k-r} b^r N^{r-1}$. Thus, we obtain

$$f(a + b \cdot N) = \sum_{k=0}^m c_k (a^k + b_k \cdot N) = f(a) + \left(\sum_{k=0}^m c_k b_k \right) \cdot N$$

Since $f(a)$ is divisible by N , we see that the right-hand side is divisible by N . Thus, so is the left-hand side.

Q 2

Given a polynomial $f(x)$ with integer coefficients so that $f(3)$ is divisible by 7 and $f(4)$ is divisible by 11. Find an integer a so that $f(a)$ is divisible by 77.

Solution: Using the result obtained in the previous question, we see that we are looking for a number $m = 3 + a \cdot 7 = 4 + b \cdot 11$. Using Q 1 we see that $f(m)$ is divisible by 7 and 11. Since 7 and 11 are co-prime, it will follow that $f(m)$ is divisible by $77 = 7 \cdot 11$.

We need to solve $4 - 3 = a \cdot 7 - b \cdot 11$. We see that $(a, b) = (8, 5)$ is a solution. Thus, $m = 3 + 8 \cdot 7 = 4 + 5 \cdot 11 = 59$.

Thus, $f(59)$ is divisible by 77. More generally, $59 + 77k$ is a solution.

Note that we do not need to know the *precise* form of $f(x)$ —only that it is a polynomial with integer coefficients.